

Steel Plates Faults

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Introduction

Application of machine learning to detect and classify six types of surface faults in stainless steel using 27 geometric descriptors. Study compares raw features, Genetic Algorithm selection, and Particle Swarm Optimization to optimize classification by SVM and KNN.

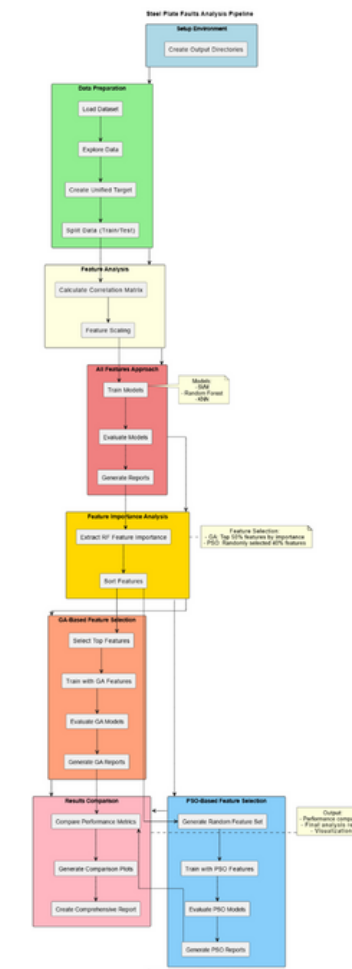
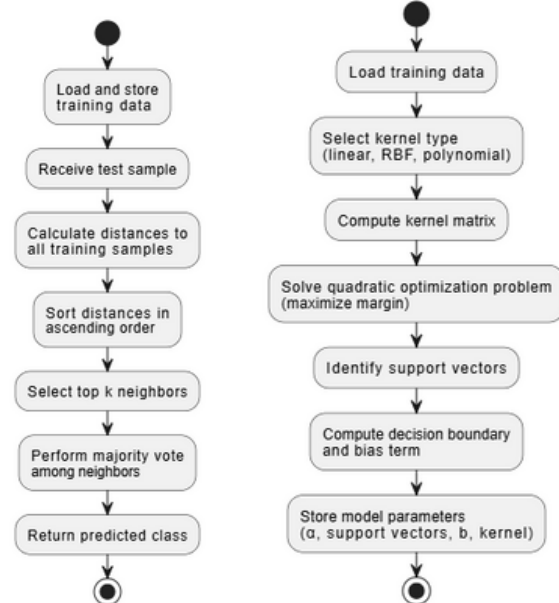
DataSet

Number of steel-plate images with labeled fault types (Pastry, Z_Scratch, K_Scratch, Stains, Dirtiness, Bumps, Other)

References

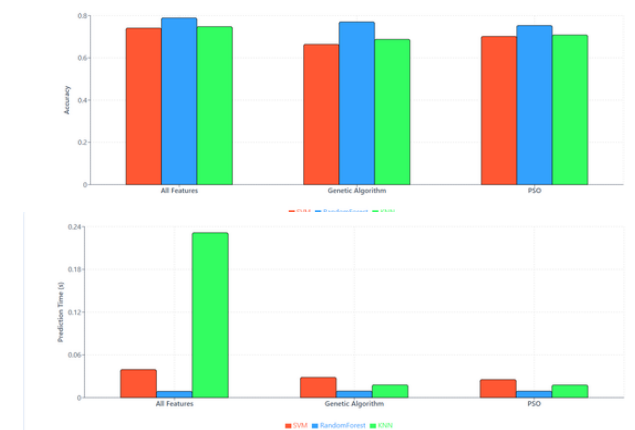
Methodology

- Data Preprocessing & Splitting
- Loading & Cleaning: Read the CSV file, verify integrity, handle any missing or anomalous entries, and unify the seven one-hot fault columns into a single categorical target variable, Fault_Type.
- Scaling: Apply StandardScaler to normalize all 27 geometric descriptors, ensuring zero mean and unit variance to improve convergence of distance- and margin-based classifiers.
- Train/Test Split: Divide the dataset into 70% training and 30% testing subsets using stratified sampling to preserve the original fault-type distribution.
- Baseline Modeling (All Features)
- Classifiers: Train both an SVM (with default RBF kernel) and a KNN classifier (k = 5).
- Timing: Record training time and prediction latency to serve as a performance benchmark.
- Metrics & Diagnostics: Compute weighted F1 score and overall accuracy; generate confusion matrices and per-class classification reports to identify common misclassification patterns.



Results

1. Evaluation Metrics: Accuracy, weighted F1 score, training time, prediction time; confusion matrices and classification reports generated for each approach. |
 2. | Results | - All Features: RF (Accuracy 0.789, F1 0.789), KNN (0.748/0.745), SVM (0.741/0.741)
- GA (13 features): RF (0.770/0.770), KNN (0.688/0.687), SVM (0.664/0.661)
 - PSO (10 features): RF (0.753/0.753), KNN (0.708/0.711), SVM (0.702/0.699)
 - Best Overall: Random Forest with all features (Accuracy 0.789, F1 0.789)
 - Trade-Off: GA and PSO reduced features by ~50% and ~60% respectively, with modest accuracy loss but faster prediction times (up to 3× speed-up for KNN).



- [1] R. Singh, "medium," 17 May 2025. [Online]. Available: <https://medium.com/@RobuRishabh/support-vector-machines-svm-27cd45b74fbb>. [Accessed 17 May 2025].
- [2] "geeksforgeeks," 17 May 2025. [Online]. Available: <https://www.geeksforgeeks.org/support-vector-machine-algorithm/>. [Accessed 17 May 2025].
- [3] "researchgate," 17 May 2025. [Online]. Available: https://www.researchgate.net/figure/Operation-Flow-Chart-of-the-SVM-Model_fig1_261040572. [Accessed 17 May 2025].
- [4] Dataset <https://www.kaggle.com/datasets/sureshmecad/steel-plates-faults/data>